

AI Career Companion Agent

for Internship Matching and Interview Preparation

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Project Documentation

1. Overview

This project is an AI-powered career assistant designed to help students and early-career professionals manage internships, improve resumes, prepare for interviews, and interact with uploaded documents through intelligent retrieval and search. The system brings together profile management, resume parsing, internship matching, document question answering, and AI-generated support in one platform.

The solution combines a Python FastAPI backend with a React frontend and uses AI techniques such as semantic similarity matching, resume analysis, and retrieval-augmented question answering. Users can sign up, upload resumes, review internship recommendations, generate cover letters, and practice interviews with personalized guidance.

Project Objectives

The primary goals of this application are to:

- simplify internship discovery based on a user profile and resume content
- extract structured information from uploaded resumes
- identify skills, experience, education, and contact information automatically
- provide useful ATS-style recommendations for resume improvement
- generate actionable interview preparation support
- enable users to ask questions about uploaded documents and receive grounded answers
- offer a secure and user-specific experience through authentication and profile storage

System Features

The application includes the following major capabilities:

- user registration, login, profile management, and password update flows
- resume upload support for PDF and DOCX files
- automatic parsing of resume sections such as summary, education, experience, skills, and projects
- skill extraction and resume profile auto-fill
- semantic internship matching using similarity search
- cover letter generation for selected internship opportunities
- ATS-based resume analysis and improvement suggestions
- mock interview preparation and interview question generation
- document upload and retrieval-based Q&A
- chatbot guidance for interview and career preparation

Technology Stack

Backend

- Python
- FastAPI
- SQLAlchemy
- SQLite
- JWT authentication
- Uvicorn

AI and Data Processing

- Sentence Transformers
- scikit-learn cosine similarity
- PyMuPDF for PDF text extraction
- python-docx for DOCX parsing
- RAG-style document retrieval workflow

Frontend

- React
- Vite
- React Router
- Axios
- Framer Motion
- react-hot-toast

Application Architecture

The system follows a layered architecture:

1. Frontend layer

The React frontend provides pages for login, registration, dashboard, resume upload, internship matching, resume analysis, interview preparation, document Q&A, and profile management. It communicates with the backend through REST API calls.

2. API layer

The FastAPI server exposes endpoints for authentication, user management, resume parsing, internships, applications, cover letter generation, document Q&A, chatbot interactions, and resume analysis.

3. Data layer

User, resume, and profile information are stored in SQLite using SQLAlchemy models. Resume and document content are saved for later retrieval and analysis.

4. AI processing layer

The backend processes uploaded documents, extracts text, identifies sections, and uses semantic similarity matching to find the best internship opportunities or document passages relevant to a question.

Resume Parsing Workflow

The resume parsing process is one of the core functions of the application. When a user uploads a resume, the system performs the following steps:

- reads the uploaded file bytes
- extracts raw text from the PDF or DOCX document
- identifies standard resume sections such as summary, education, skills, experience, and projects
- extracts contact details such as email and phone number
- detects technical skills using a predefined skill dictionary
- stores the parsed result in the user profile and resume database
- uses the extracted skills to query internships and find the best matches

The parser is designed to work with resumes that use conventional section headings. This improves extraction accuracy and ensures that the profile data remains consistent and reusable.

Recommended Resume Structure

For best results, a resume should use a clear and consistent structure such as:

- Contact Information
- Summary or Professional Profile
- Skills
- Education
- Experience or Internships
- Projects
- Certifications

This structure matches the parser's expected logic and improves both extraction quality and profile auto-fill accuracy.

Document Q&A Workflow

The document Q&A feature supports non-resume document uploads. The system follows a retrieval-based workflow:

- upload a file
- extract text from the document
- split the content into overlapping chunks
- retrieve the most relevant chunks for the user question
- generate an answer based on the retrieved content

This approach helps answer questions from long files more accurately by focusing on the most relevant passages instead of processing the whole document as one block.

Internship Match Process

When a resume is parsed, the extracted skills are combined with the resume text and compared against an internship dataset. This uses semantic matching and similarity scoring to identify the internship opportunities that best fit the candidate profile. The scoring system returns ranked results and highlights the matching skills for each listed opportunity.

User Profile Auto-Fill

One important feature of the application is that parsed resume content automatically updates the user's profile. The system captures details such as:

- phone number
- summary
- extracted skills

- education
- internship or work experience
- projects
- certifications
- email address from resume content

This ensures that later features such as cover letters, interview prep, and dashboard content have consistent user data without manual re-entry.

Resume Analysis and Feedback

The resume analyzer checks the uploaded resume for ATS compatibility and content quality. It evaluates whether the user has included and structured key sections correctly and offers feedback on areas such as:

- contact information completeness
- skills relevance
- experience detail
- education clarity
- formatting and structure
- ATS keyword coverage

This helps users improve their resumes before applying to internships or jobs.

Security and Data Handling

The application stores user-specific data in a local SQLite database and uses authentication tokens for protected endpoints. For practical deployment, it is recommended to:

- use a secure secret key for JWT signing
- define trusted CORS origins carefully
- avoid uploading confidential files to an untrusted deployment
- keep production setups separate from local development environments

Project Folder Structure

The project is organized as follows:

- `internship_rag_system.py`: main FastAPI application, API routes, models, parser, and AI logic
- `tokenization_rag.py`: retrieval and tokenization related utilities
- `README.md`: setup, usage, and project overview documentation
- `frontend/`: React application with pages, components, API client, and context providers
- `airesume.db`: local SQLite database created during application startup
- `LICENSE`: project license information

Operational Workflow

A typical user journey is:

- 1. create an account and log in**
- 2. upload a resume in PDF or DOCX format**
- 3. parse and analyze the resume automatically**
- 4. review internship job matches and recommended opportunities**
- 5. generate a tailored cover letter**
- 6. use the resume analyzer for improvement suggestions**

7. practice technical and HR interview questions

8. ask questions about uploaded documents using the Q&A feature

Conclusion

This project demonstrates how AI and retrieval-based systems can support the job-search and career-preparation process in a practical and user-friendly way. The combination of resume parsing, semantic matching, interview support, and document retrieval creates a single digital assistant that helps students and early-career professionals take a more structured and informed approach to internships and career growth.